

Before the talk:

Open these websites:

[http://sourceforge.net/apps/trac/groupeexplorer/wiki/
The First Five Symmetric Groups/](http://sourceforge.net/apps/trac/groupeexplorer/wiki/The+First+Five+Symmetric+Groups/)

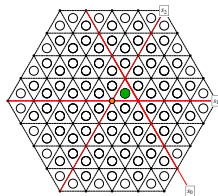
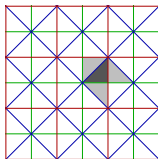
Bring Zome Tools:

Permutohedron

Affine A_3

A combinatorial introduction to reflection groups

Christopher R. H. Hanusa
Queens College, CUNY



people.qc.cuny.edu/chanusa > Talks

Groups

Today, we will discuss the combinatorics of **groups**.

- ▶ Made up of a set of elements $W = \{w_1, w_2, \dots\}$.
- ▶ Multiplication of two elements $w_1 w_2$ stays in the group.

Groups

Today, we will discuss the combinatorics of **groups**.

- ▶ Made up of a set of elements $W = \{w_1, w_2, \dots\}$.
- ▶ Multiplication of two elements $w_1 w_2$ stays in the group.
 - ▶ ALTHOUGH, it might **not** be the case that $w_1 w_2 = w_2 w_1$.

Groups

Today, we will discuss the combinatorics of **groups**.

- ▶ Made up of a set of elements $W = \{w_1, w_2, \dots\}$.
- ▶ Multiplication of two elements $w_1 w_2$ stays in the group.
 - ▶ ALTHOUGH, it might **not** be the case that $w_1 w_2 = w_2 w_1$.
- ▶ There is an identity element (**id**) & Every element has an inverse.

Groups

Today, we will discuss the combinatorics of **groups**.

- ▶ Made up of a set of elements $W = \{w_1, w_2, \dots\}$.
- ▶ Multiplication of two elements $w_1 w_2$ stays in the group.
 - ▶ ALTHOUGH, it might **not** be the case that $w_1 w_2 = w_2 w_1$.
- ▶ There is an identity element (**id**) & Every element has an inverse.
- ▶ *Group elements take on the role of both objects and functions.*

Groups

Today, we will discuss the combinatorics of **groups**.

- ▶ Made up of a set of elements $W = \{w_1, w_2, \dots\}$.
- ▶ Multiplication of two elements $w_1 w_2$ stays in the group.
 - ▶ ALTHOUGH, it might **not** be the case that $w_1 w_2 = w_2 w_1$.
- ▶ There is an identity element (**id**) & Every element has an inverse.
- ▶ *Group elements take on the role of both objects and functions.*

(Non-zero real numbers)

- ▶ We can multiply a and b
- ▶ It is the case that $ab = ba$
- ▶ **1** is the identity: $a \cdot 1 = a$
- ▶ The inverse of a is $1/a$.

Groups

Today, we will discuss the combinatorics of **groups**.

- ▶ Made up of a set of elements $W = \{w_1, w_2, \dots\}$.
- ▶ Multiplication of two elements $w_1 w_2$ stays in the group.
 - ▶ ALTHOUGH, it might **not** be the case that $w_1 w_2 = w_2 w_1$.
- ▶ There is an identity element (**id**) & Every element has an inverse.
- ▶ *Group elements take on the role of both objects and functions.*

(Non-zero real numbers)

- ▶ We can multiply a and b
- ▶ It is the case that $ab = ba$
- ▶ **1** is the identity: $a \cdot 1 = a$
- ▶ The inverse of a is $1/a$.

(Invertible $n \times n$ matrices.)

- ▶ We can multiply A and B
- ▶ *Rarely* is $AB = BA$
- ▶ I_n is the identity: $A \cdot I_n = A$
- ▶ The inverse of A exists: A^{-1} .

Reflection Groups

More specifically, we will discuss **reflection groups** W .

- ▶ W is generated by a set of **generators** $S = \{s_1, s_2, \dots, s_k\}$.
- ▶ Along with a set of **relations**.

Reflection Groups

More specifically, we will discuss **reflection groups** W .

- ▶ W is generated by a set of **generators** $S = \{s_1, s_2, \dots, s_k\}$.
 - ▶ Every $w \in W$ can be written as a product of generators.
- ▶ Along with a set of **relations**.

Reflection Groups

More specifically, we will discuss **reflection groups** W .

- ▶ W is generated by a set of **generators** $S = \{s_1, s_2, \dots, s_k\}$.
 - ▶ Every $w \in W$ can be written as a product of generators.
- ▶ Along with a set of **relations**.
 - ▶ These are rules to convert between expressions.
 - ▶ $s_i^2 = \text{id}$. ~~and~~ $(s_i s_j)^{\text{power}} = \text{id}$. (Write down)

Reflection Groups

More specifically, we will discuss **reflection groups** W .

- ▶ W is generated by a set of **generators** $S = \{s_1, s_2, \dots, s_k\}$.
 - ▶ Every $w \in W$ can be written as a product of generators.
- ▶ Along with a set of **relations**.
 - ▶ These are rules to convert between expressions.
 - ▶ $s_i^2 = \text{id}$. ~~and~~ $(s_i s_j)^{\text{power}} = \text{id}$. (Write down)

For example, $w = s_3 s_2 s_1 s_1 s_2 s_4 = s_3 s_2 \text{id} s_2 s_4 = s_3 \text{id} s_4 = s_3 s_4$

Reflection Groups

More specifically, we will discuss **reflection groups** W .

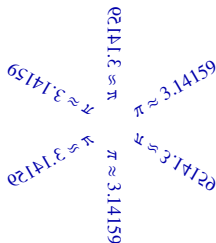
- ▶ W is generated by a set of **generators** $S = \{s_1, s_2, \dots, s_k\}$.
 - ▶ Every $w \in W$ can be written as a product of generators.
- ▶ Along with a set of **relations**.
 - ▶ These are rules to convert between expressions.
 - ▶ $s_i^2 = \text{id}$. ~~and~~ $(s_i s_j)^{\text{power}} = \text{id}$. (Write down)

For example, $w = s_3 s_2 s_1 s_1 s_2 s_4 = s_3 s_2 \text{id} s_2 s_4 = s_3 \text{id} s_4 = s_3 s_4$

Why should we use **these** rules?

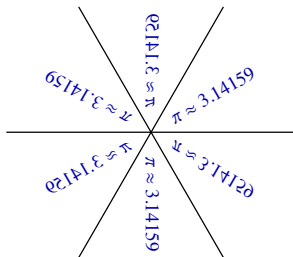
Pi in the cold of winter

Behold: The perfect wallpaper design for math majors:



Pi in the cold of winter

Behold: The perfect wallpaper design for math majors:

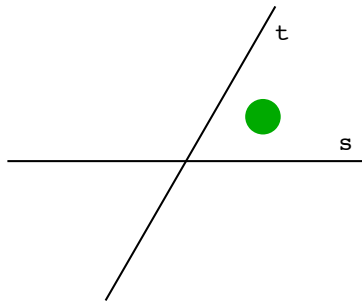


To see the reflections, we insert some **hyperplanes** that act as mirrors.

- ▶ In two dimensions, a hyperplane is simply a line.
- ▶ In three dimensions, a hyperplane is a plane.

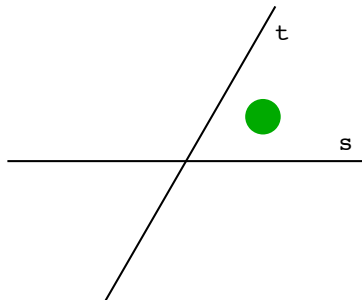
Reflection Groups

- These regions can be thought of as group elements. Place id .



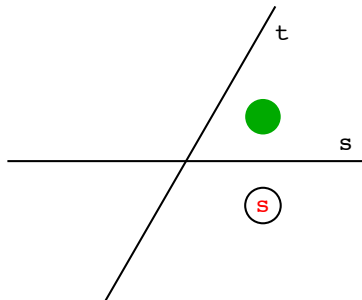
Reflection Groups

- ▶ These regions can be thought of as group elements. Place id .
- ▶ The action of multiplying (on the left) by a generator s corresponds to a reflection across a hyperplane H_s . ($s^2 = \text{id}$)



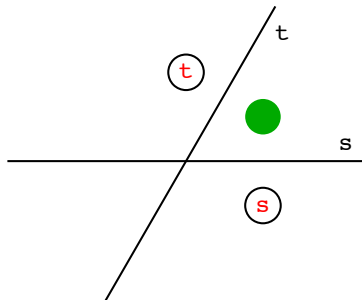
Reflection Groups

- ▶ These regions can be thought of as group elements. Place id .
- ▶ The action of multiplying (on the left) by a generator s corresponds to a reflection across a hyperplane H_s . ($s^2 = \text{id}$)



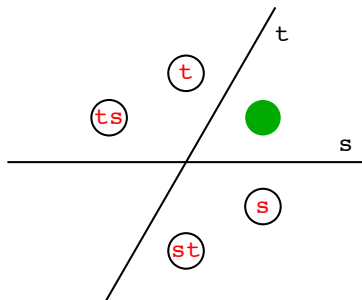
Reflection Groups

- ▶ These regions can be thought of as group elements. Place id .
- ▶ The action of multiplying (on the left) by a generator s corresponds to a reflection across a hyperplane H_s . ($s^2 = \text{id}$)



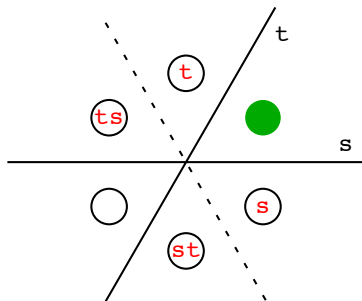
Reflection Groups

- ▶ These regions can be thought of as group elements. Place id .
- ▶ The action of multiplying (on the left) by a generator s corresponds to a reflection across a hyperplane H_s . ($s^2 = \text{id}$)



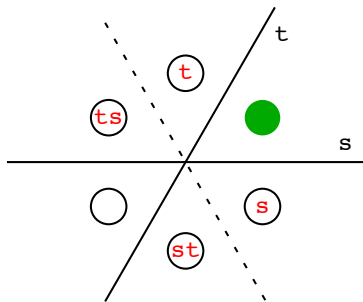
Reflection Groups

- ▶ These regions can be thought of as group elements. Place **id**.
- ▶ The action of multiplying (on the left) by a generator s corresponds to a reflection across a hyperplane H_s . ($s^2 = \text{id}$)



Reflection Groups

- ▶ These regions can be thought of as group elements. Place **id**.
- ▶ The action of multiplying (on the left) by a generator s corresponds to a reflection across a hyperplane H_s . ($s^2 = \text{id}$)

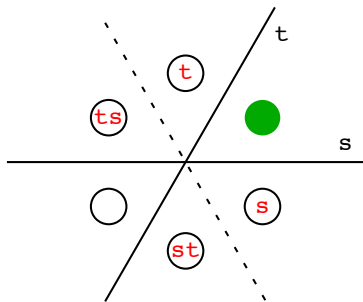


We see:

$$\text{▶ } sts = tst \leftrightarrow ststst = ts\textcolor{red}{t}tst$$

Reflection Groups

- ▶ These regions can be thought of as group elements. Place **id**.
- ▶ The action of multiplying (on the left) by a generator s corresponds to a reflection across a hyperplane H_s . ($s^2 = \text{id}$)

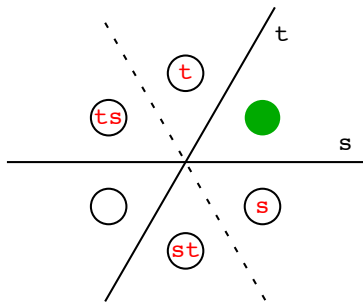


We see:

$$\text{▶ } sts = tst \leftrightarrow ststst = t\textcolor{blue}{s}st$$

Reflection Groups

- ▶ These regions can be thought of as group elements. Place **id**.
- ▶ The action of multiplying (on the left) by a generator s corresponds to a reflection across a hyperplane H_s . ($s^2 = \text{id}$)

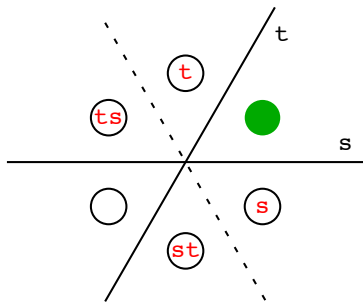


We see:

$$\triangleright sts = tst \leftrightarrow ststst = tt$$

Reflection Groups

- ▶ These regions can be thought of as group elements. Place **id**.
- ▶ The action of multiplying (on the left) by a generator s corresponds to a reflection across a hyperplane H_s . ($s^2 = \text{id}$)

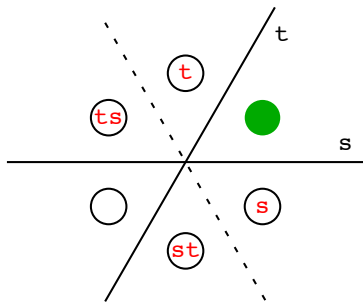


We see:

- ▶ $sts = tst \leftrightarrow ststst = \text{id}$
Shows $(st)^3 = \text{id}$ is natural.

Reflection Groups

- ▶ These regions can be thought of as group elements. Place **id**.
- ▶ The action of multiplying (on the left) by a generator s corresponds to a reflection across a hyperplane H_s . ($s^2 = \text{id}$)

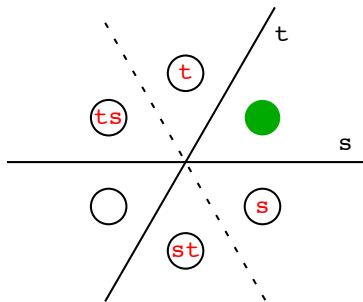


We see:

- ▶ $sts = tst \leftrightarrow ststst = \text{id}$
Shows $(st)^3 = \text{id}$ is natural.
- ▶ Our group has six elements:
 $\{\text{id}, s, t, st, ts, sts\}$.

Reflection Groups

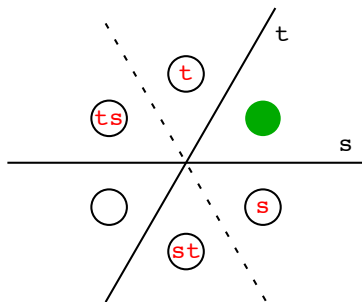
- ▶ These regions can be thought of as group elements. Place **id**.
- ▶ The action of multiplying (on the left) by a generator s corresponds to a reflection across a hyperplane H_s . ($s^2 = \text{id}$)



We see:

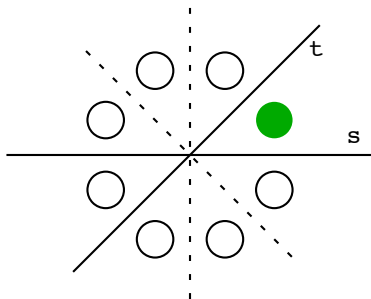
- ▶ $sts = tst \leftrightarrow ststst = \text{id}$
Shows $(st)^3 = \text{id}$ is natural.
- ▶ Our group has six elements: $\{\text{id}, s, t, st, ts, sts\}$.
- ▶ This is the group of symmetries of a hexagon.

Reflection Groups



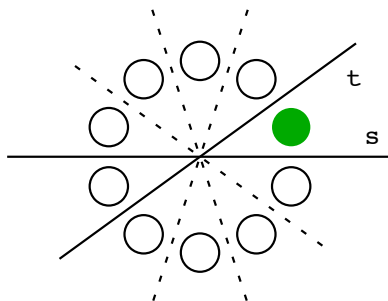
- ▶ When the angle between H_s and H_t is $\frac{\pi}{3}$, relation is $(st)^3 = \text{id}$.
- ▶ The size of the group is $|S| = 6$.

Reflection Groups



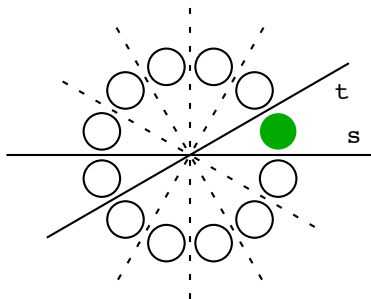
- ▶ When the angle between H_s and H_t is $\frac{\pi}{4}$, relation is $(st)^4 = \text{id}$.
- ▶ The size of the group is $|S| = 8$.

Reflection Groups



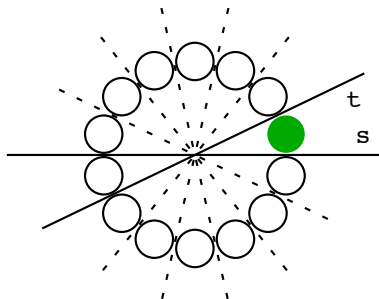
- ▶ When the angle between H_s and H_t is $\frac{\pi}{5}$, relation is $(st)^5 = \text{id}$.
- ▶ The size of the group is $|S| = 10$.

Reflection Groups



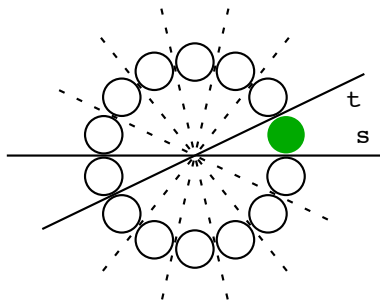
- ▶ When the angle between H_s and H_t is $\frac{\pi}{6}$, relation is $(st)^6 = \text{id}$.
- ▶ The size of the group is $|S| = 12$.

Reflection Groups



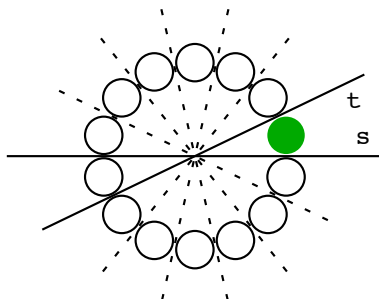
- ▶ When the angle between H_s and H_t is $\frac{\pi}{n}$, relation is $(st)^n = \text{id}$.
- ▶ The size of the group is $|S| = 2n$.

Reflection Groups



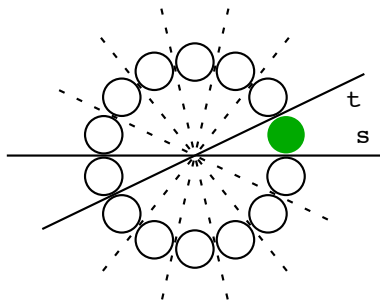
- ▶ When the angle between H_s and H_t is $\frac{\pi}{n}$, relation is $(st)^n = \text{id}$.
- ▶ The size of the group is $|S| = 2n$.
- ▶ All finite reflection groups in the plane are these **dihedral groups**.

Reflection Groups



- ▶ When the angle between H_s and H_t is $\frac{\pi}{n}$, relation is $(st)^n = \text{id}$.
- ▶ The size of the group is $|S| = 2n$.
- ▶ All finite reflection groups in the plane are these **dihedral groups**.
- ▶ **Two directions:** infinite and higher dimensional.

Reflection Groups



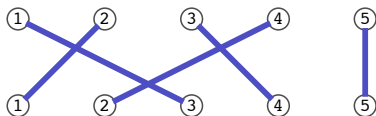
- ▶ When the angle between H_s and H_t is $\frac{\pi}{n}$, relation is $(st)^n = \text{id}$.
- ▶ The size of the group is $|S| = 2n$.
- ▶ All finite reflection groups in the plane are these **dihedral groups**.
- ▶ **Two directions:** infinite and **higher dimensional**.

Permutations are a group

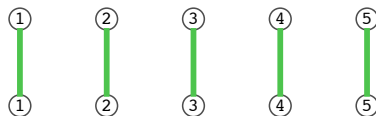
An n -**permutation** is a permutation of $\{1, 2, \dots, n\}$.

- Write in **one-line notation** or use a **string diagram**:

31425



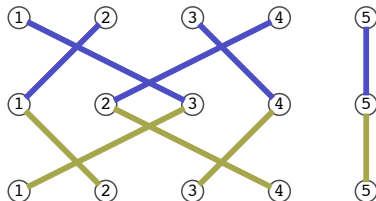
12345



n -Permutations form
the **Symmetric group** S_n .

- We can multiply permutations.
- The identity permutation is **id** = **1234...n**.
- Inverse permutation: Flip the string diagram upside down!

31425 $\times$ **24135** = **12345**

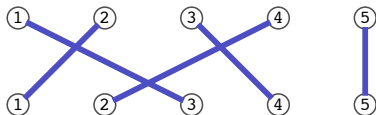


Permutations are a group

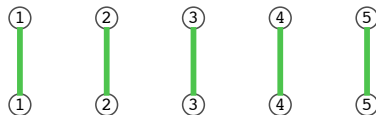
An n -**permutation** is a permutation of $\{1, 2, \dots, n\}$.

- Write in **one-line notation** or use a **string diagram**:

31425



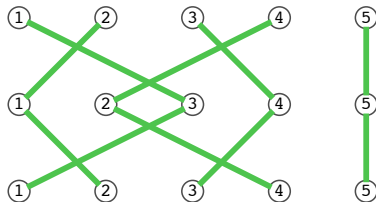
12345



n -Permutations form
the **Symmetric group** S_n .

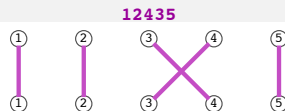
- We can multiply permutations.
- The identity permutation is $\text{id} = 1234\dots n$.
- Inverse permutation: Flip the string diagram upside down!

31425 $\times$ **24135** = **12345**



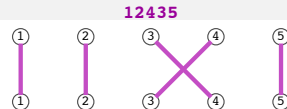
Permutations as a reflection group

A special type of permutation is an **adjacent transposition**, switching two adjacent entries.



Permutations as a reflection group

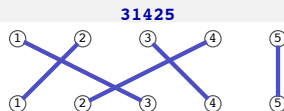
A special type of permutation is an **adjacent transposition**, switching two adjacent entries.



- Write $s_i : (i) \leftrightarrow (i + 1)$. (e.g. $s_3 = 12435$).

Permutations as a reflection group

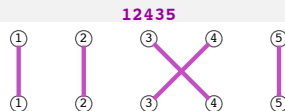
A special type of permutation is an **adjacent transposition**, switching two adjacent entries.



- ▶ Write $s_i : (i) \leftrightarrow (i + 1)$. (e.g. $s_3 = 12435$).
- ★ Every n -permutation is a product of adjacent transpositions.
 - ▶ (Construct any string diagram through individual twists.)
 - ▶ *Example.* Write 31425 as $s_1 s_3 s_2$.

Permutations as a reflection group

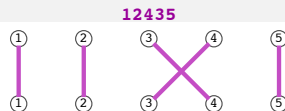
A special type of permutation is an **adjacent transposition**, switching two adjacent entries.



- ▶ Write $s_i : (i) \leftrightarrow (i + 1)$. (e.g. $s_3 = 12435$).
- ★ Every n -permutation is a product of adjacent transpositions.
 - ▶ (Construct any string diagram through individual twists.)
 - ▶ *Example.* Write 31425 as $s_1 s_3 s_2$.
- ▶ $S = \{s_1, s_2, \dots, s_{n-1}\}$ are **generators** of S_n .

Permutations as a reflection group

A special type of permutation is an **adjacent transposition**, switching two adjacent entries.

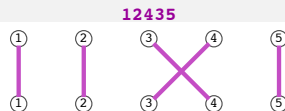


- ▶ Write $s_i : (i) \leftrightarrow (i + 1)$. (e.g. $s_3 = 1\,2\,4\,3\,5$).
- ★ Every n -permutation is a product of adjacent transpositions.
 - ▶ (Construct any string diagram through individual twists.)
 - ▶ *Example.* Write $3\,1\,4\,2\,5$ as $s_1 s_3 s_2$.
- ▶ $S = \{s_1, s_2, \dots, s_{n-1}\}$ are **generators** of S_n .

A reflection group also has relations:

Permutations as a reflection group

A special type of permutation is an **adjacent transposition**, switching two adjacent entries.



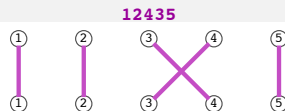
- ▶ Write $s_i : (i) \leftrightarrow (i + 1)$. (e.g. $s_3 = 12435$).
- ★ Every n -permutation is a product of adjacent transpositions.
 - ▶ (Construct any string diagram through individual twists.)
 - ▶ *Example.* Write 31425 as $s_1 s_3 s_2$.
- ▶ $S = \{s_1, s_2, \dots, s_{n-1}\}$ are **generators** of S_n .

A reflection group also has relations:

- ▶ First, $s_i^2 = \text{id}$.

Permutations as a reflection group

A special type of permutation is an **adjacent transposition**, switching two adjacent entries.



► Write $s_i : (i) \leftrightarrow (i + 1)$. (e.g. $s_3 = 1\,2\,4\,3\,5$).

★ Every n -permutation is a product of adjacent transpositions.

► (Construct any string diagram through individual twists.)

► *Example.* Write $3\,1\,4\,2\,5$ as $s_1 s_3 s_2$.

► $S = \{s_1, s_2, \dots, s_{n-1}\}$ are **generators** of S_n .

A reflection group also has relations:

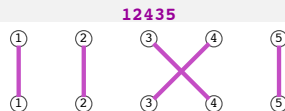
► First, $s_i^2 = \text{id}$.

► Consecutive generators don't commute: $s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1}$

12345	12345
2 1345	1 3 245
2 3 145	3 1 245
3 2 145	3 2 145

Permutations as a reflection group

A special type of permutation is an **adjacent transposition**, switching two adjacent entries.



- ▶ Write $s_i : (i) \leftrightarrow (i + 1)$. (e.g. $s_3 = 1\ 2\ 4\ 3\ 5$).
- ★ Every n -permutation is a product of adjacent transpositions.
 - ▶ (Construct any string diagram through individual twists.)
 - ▶ *Example.* Write $3\ 1\ 4\ 2\ 5$ as $s_1 s_3 s_2$.
 - ▶ $S = \{s_1, s_2, \dots, s_{n-1}\}$ are **generators** of S_n .

A reflection group also has relations:

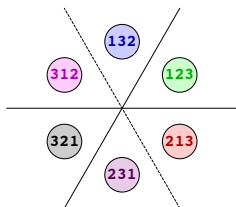
- ▶ First, $s_i^2 = \text{id}$.
- ▶ Consecutive generators don't commute: $s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1}$
- ▶ Non-consecutive generators DO commute: $s_i s_j = s_j s_i$

12345	12345
2 1345	1 3 245
2 3 145	3 1245
3 2145	3 2 145

21345

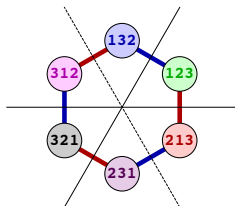
Visualizing symmetric groups

We have already seen S_3 , generated by $\{s_1, s_2\}$:



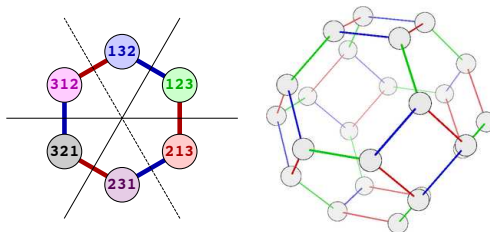
Visualizing symmetric groups

We have already seen S_3 , generated by $\{s_1, s_2\}$:



Visualizing symmetric groups

We have already seen S_3 , generated by $\{s_1, s_2\}$:

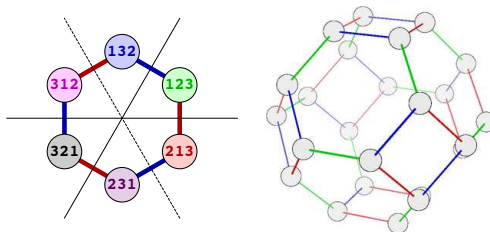


We can visualize S_4 as a **permutohedron**, generated by $\{s_1, s_2, s_3\}$.

[sourceforge.net/apps/trac/grouperexplorer/wiki/The First Five Symmetric Groups/](https://sourceforge.net/apps/trac/grouperexplorer/wiki/The%20First%20Five%20Symmetric%20Groups/)

Visualizing symmetric groups

We have already seen S_3 , generated by $\{s_1, s_2\}$:



We can visualize S_4 as a **permutohedron**, generated by $\{s_1, s_2, s_3\}$.

[sourceforge.net/apps/trac/grouperexplorer/wiki/The First Five Symmetric Groups/](http://sourceforge.net/apps/trac/grouperexplorer/wiki/The%20First%20Five%20Symmetric%20Groups/)

They also give a way to see $S_5 \dots$

Higher-dimension symmetric groups

How can we “see” a reflection group in higher dimensions?

The relation $(s_i s_j)^m$ determines the angle between hyperplanes H_i, H_j :

$$\triangleright (s_i s_j)^2 = \text{id} \iff \theta(H_i, H_j) = \pi/2$$

$$\triangleright (s_i s_j)^3 = \text{id} \iff \theta(H_i, H_j) = \pi/3$$

For S_6 , we expect an angle of 60° between the hyperplane pairs

$$(H_1, H_2), (H_2, H_3), (H_3, H_4), \text{ and } (H_4, H_5).$$

Every other pair will be perpendicular.

All finite reflection groups

Or see with a **Coxeter diagram**:

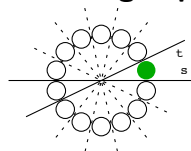
- ▶ **Vertices**: One for every generator i
- ▶ **Edges**: Between i and j when $m_{i,j} \geq 3$.
Label edges with $m_{i,j}$ when ≥ 4 .

All finite reflection groups

Or see with a **Coxeter diagram**:

- ▶ **Vertices:** One for every generator i
- ▶ **Edges:** Between i and j when $m_{i,j} \geq 3$.
Label edges with $m_{i,j}$ when ≥ 4 .

Dihedral groups



Generators: s and t .

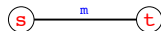
Relation: $(st)^m = \text{id}$

All finite reflection groups

Or see with a **Coxeter diagram**:

- ▶ **Vertices**: One for every generator i
- ▶ **Edges**: Between i and j when $m_{i,j} \geq 3$.
Label edges with $m_{i,j}$ when ≥ 4 .

Dihedral groups



Generators: s and t .

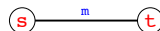
Relation: $(st)^m = \text{id}$

All finite reflection groups

Or see with a **Coxeter diagram**:

- **Vertices**: One for every generator i
- **Edges**: Between i and j when $m_{i,j} \geq 3$.
Label edges with $m_{i,j}$ when ≥ 4 .

Dihedral groups



Generators: s and t .
Relation: $(st)^m = \text{id}$

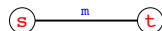
Symmetric groups:
 A Coxeter diagram for a symmetric group, consisting of a path of vertices labeled $1, 2, \dots, n$ connected by edges.

All finite reflection groups

Or see with a **Coxeter diagram**:

- **Vertices**: One for every generator i
- **Edges**: Between i and j when $m_{i,j} \geq 3$.
Label edges with $m_{i,j}$ when ≥ 4 .

Dihedral groups

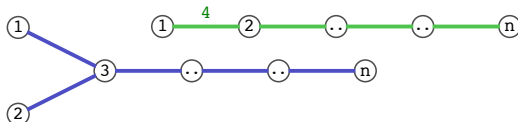


Generators: s and t .
Relation: $(st)^m = \text{id}$

Symmetric groups:



Infinite families:

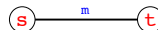


All finite reflection groups

Or see with a **Coxeter diagram**:

- **Vertices**: One for every generator i
- **Edges**: Between i and j when $m_{i,j} \geq 3$.
Label edges with $m_{i,j}$ when ≥ 4 .

Dihedral groups

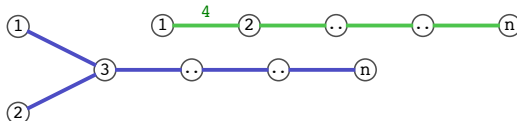


Generators: s and t .
Relation: $(st)^m = \text{id}$

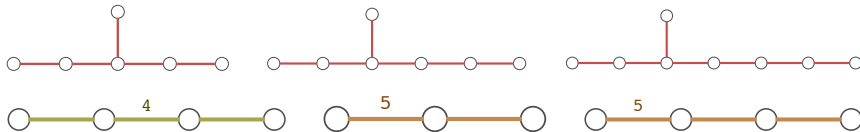
Symmetric groups:



Infinite families:



Exceptional types:



Wallpaper Groups

The art of M. C. Escher plays upon symmetries in the plane.



Wallpaper Groups

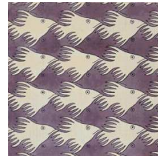
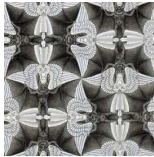
The art of M. C. Escher plays upon symmetries in the plane.



An **isometry** of the plane is a transformation that preserves distance.

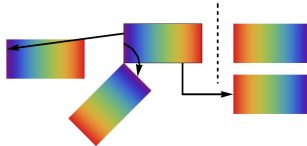
Wallpaper Groups

The art of M. C. Escher plays upon symmetries in the plane.



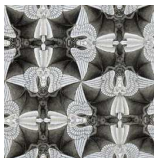
An **isometry** of the plane is a transformation that preserves distance.

Think: translations, rotations, reflections, glide reflections.



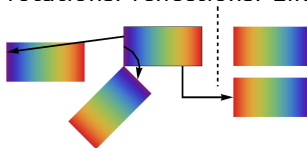
Wallpaper Groups

The art of M. C. Escher plays upon symmetries in the plane.



An **isometry** of the plane is a transformation that preserves distance.

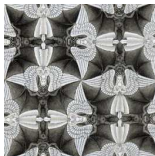
Think: translations, rotations, reflections, glide reflections.



A **wallpaper group** is a group of isometries of the plane with two independent translations.

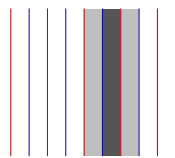
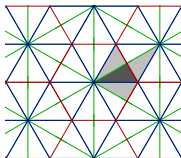
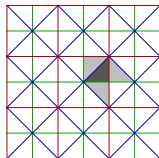
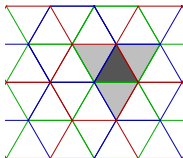
Wallpaper Groups

The art of M. C. Escher plays upon symmetries in the plane.



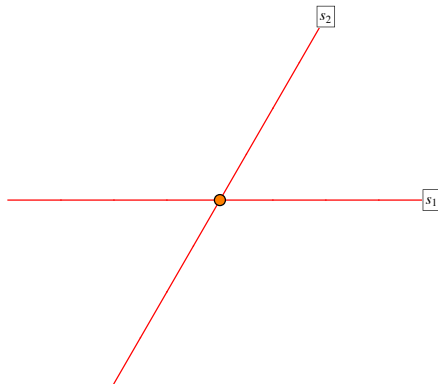
An **isometry** of the plane is a transformation that preserves distance.

A **wallpaper group** is a group of isometries of the plane with two independent translations. Some are also reflection groups:



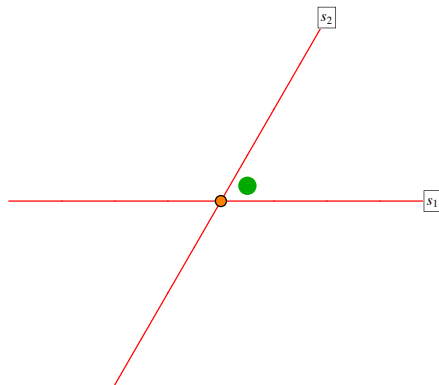
Infinite Reflection Groups

Constructing an infinite reflection group: the affine permutations $\tilde{S}_n$.



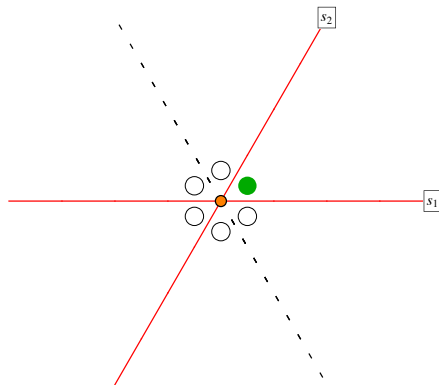
Infinite Reflection Groups

Constructing an infinite reflection group: the affine permutations $\tilde{S}_n$.



Infinite Reflection Groups

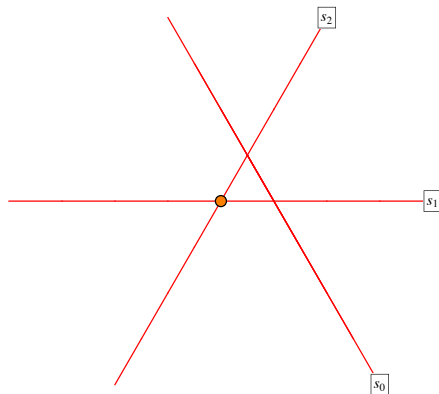
Constructing an infinite reflection group: the affine permutations $\tilde{S}_n$.



Infinite Reflection Groups

Constructing an infinite reflection group: the **affine permutations** $\tilde{S}_n$.

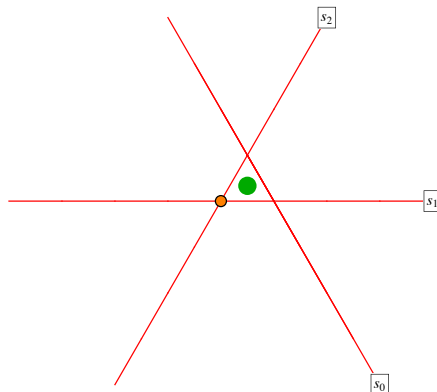
- Add a new generator s_0 and a new **affine** hyperplane H_0 .



Infinite Reflection Groups

Constructing an infinite reflection group: the **affine permutations** $\tilde{S}_n$.

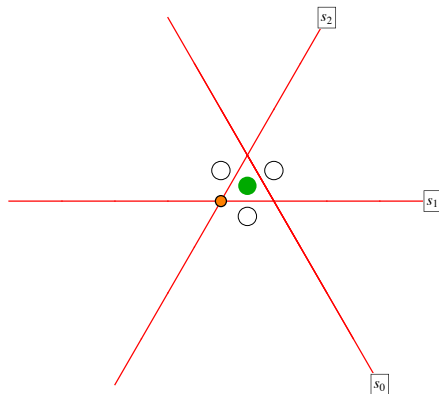
- Add a new generator s_0 and a new **affine** hyperplane H_0 .



Infinite Reflection Groups

Constructing an infinite reflection group: the **affine permutations** $\tilde{S}_n$.

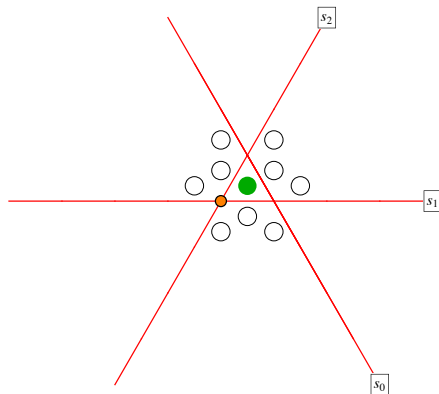
- Add a new generator s_0 and a new **affine** hyperplane H_0 .



Infinite Reflection Groups

Constructing an infinite reflection group: the **affine permutations** $\tilde{S}_n$.

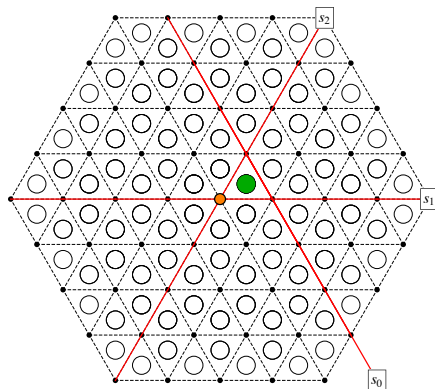
- Add a new generator s_0 and a new **affine** hyperplane H_0 .



Infinite Reflection Groups

Constructing an infinite reflection group: the affine permutations $\tilde{S}_n$.

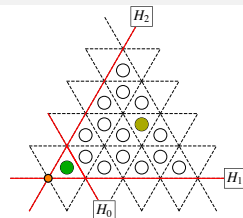
- Add a new generator s_0 and a new **affine** hyperplane H_0 .



Elements generated by $\{s_0, s_1, s_2\}$ correspond to **alcoves** here.

Combinatorics of affine permutations

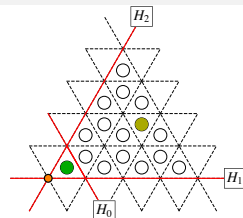
Many ways to reference elements in $\tilde{S}_n$.



Combinatorics of affine permutations

Many ways to reference elements in $\tilde{S}_n$.

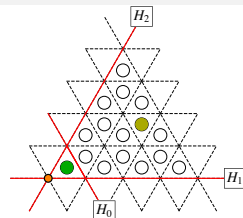
- **Geometry.** Point to the alcove.



Combinatorics of affine permutations

Many ways to reference elements in $\tilde{S}_n$.

- **Geometry.** Point to the alcove.
- **Alcove coordinates.** Keep track of how many hyperplanes of each type you have crossed to get to your alcove.



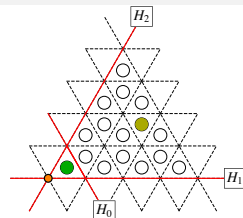
Coordinates:

3	1
1	

Combinatorics of affine permutations

Many ways to reference elements in $\tilde{S}_n$.

- **Geometry.** Point to the alcove.
- **Alcove coordinates.** Keep track of how many hyperplanes of each type you have crossed to get to your alcove.
- **Word.** Write the element as a (short) product of generators.



Coordinates:

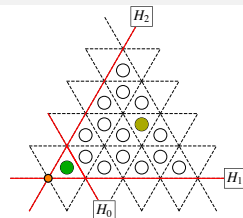
3	1
1	

Word: $s_0 s_1 s_2 s_1 s_0$

Combinatorics of affine permutations

Many ways to reference elements in $\tilde{S}_n$.

- ▶ **Geometry.** Point to the alcove.
- ▶ **Alcove coordinates.** Keep track of how many hyperplanes of each type you have crossed to get to your alcove.
- ▶ **Word.** Write the element as a (short) product of generators.
- ▶ **One-line notation.** Similar to writing finite permutations as 312.



Coordinates:

3	1
1	

Word: $s_0 s_1 s_2 s_1 s_0$

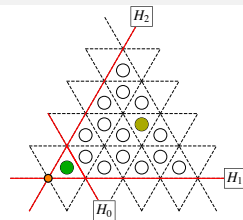
Permutation:

$(-3, 2, 7)$

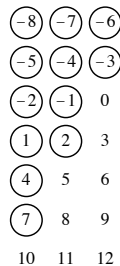
Combinatorics of affine permutations

Many ways to reference elements in $\tilde{S}_n$.

- ▶ **Geometry.** Point to the alcove.
- ▶ **Alcove coordinates.** Keep track of how many hyperplanes of each type you have crossed to get to your alcove.
- ▶ **Word.** Write the element as a (short) product of generators.
- ▶ **One-line notation.** Similar to writing finite permutations as 312.
- ▶ **Abacus diagram.** Columns of numbers.



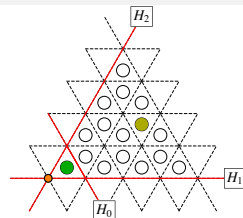
Abacus diagram:



Combinatorics of affine permutations

Many ways to reference elements in $\tilde{S}_n$.

- ▶ **Geometry.** Point to the alcove.
- ▶ **Alcove coordinates.** Keep track of how many hyperplanes of each type you have crossed to get to your alcove.
- ▶ **Word.** Write the element as a (short) product of generators.
- ▶ **One-line notation.** Similar to writing finite permutations as 312.
- ▶ **Abacus diagram.** Columns of numbers.
- ▶ **Core partition.** Hook length condition.



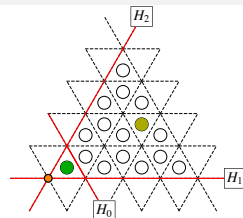
Core partition:

0	1	2	0
2	0		
1			
0			

Combinatorics of affine permutations

Many ways to reference elements in $\tilde{S}_n$.

- ▶ **Geometry.** Point to the alcove.
- ▶ **Alcove coordinates.** Keep track of how many hyperplanes of each type you have crossed to get to your alcove.
- ▶ **Word.** Write the element as a (short) product of generators.
- ▶ **One-line notation.** Similar to writing finite permutations as 312.
- ▶ **Abacus diagram.** Columns of numbers.
- ▶ **Core partition.** Hook length condition.
- ▶ **Bounded partition.** Part size bounded.



Core partition:

0	1	2	0
2	0		
1			
0			

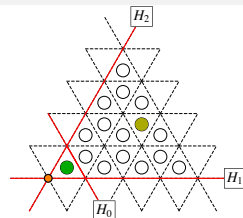
Bounded partition:

0	1
2	
1	
0	

Combinatorics of affine permutations

Many ways to reference elements in $\tilde{S}_n$.

- ▶ **Geometry.** Point to the alcove.
- ▶ **Alcove coordinates.** Keep track of how many hyperplanes of each type you have crossed to get to your alcove.
- ▶ **Word.** Write the element as a (short) product of generators.
- ▶ **One-line notation.** Similar to writing finite permutations as 312.
- ▶ **Abacus diagram.** Columns of numbers.
- ▶ **Core partition.** Hook length condition.
- ▶ **Bounded partition.** Part size bounded.
- ▶ **Others!** Lattice path, order ideal, etc.

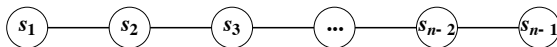


They all play nicely with each other.

Affine permutations

(Finite) n -Permutations S_n

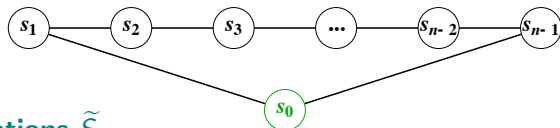
► Visually:



Affine permutations

(Finite) n -Permutations S_n

► Visually:



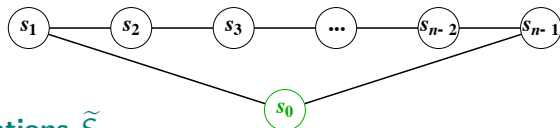
Affine n -Permutations $\tilde{S}_n$

- Generators: $\{s_0, s_1, \dots, s_{n-1}\}$
- s_0 has a braid relation with s_1 and s_{n-1}
- How does this impact **one-line notation**?

Affine permutations

(Finite) n -Permutations S_n

- Visually:



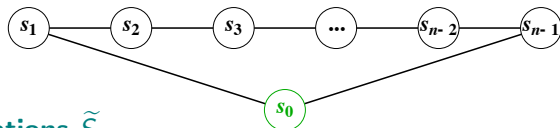
Affine n -Permutations $\tilde{S}_n$

- Generators: $\{s_0, s_1, \dots, s_{n-1}\}$
- s_0 has a braid relation with s_1 and s_{n-1}
- How does this impact **one-line notation**?
 - Perhaps interchanges 1 and n ?

Affine permutations

(Finite) n -Permutations S_n

► Visually:



Affine n -Permutations $\tilde{S}_n$

- Generators: $\{s_0, s_1, \dots, s_{n-1}\}$
- s_0 has a braid relation with s_1 and s_{n-1}
- How does this impact **one-line notation**?
 - Perhaps interchanges 1 and n ?
 - Not quite! (Would add a relation.)

Window notation

Affine n -Permutations $\tilde{S}_n$ (G. Lusztig 1983, H. Eriksson, 1994)

Write an element $\tilde{w} \in \tilde{S}_n$ in 1-line notation as a permutation of $\mathbb{Z}$.

Window notation

Affine n -Permutations $\tilde{S}_n$ (G. Lusztig 1983, H. Eriksson, 1994)

Write an element $\tilde{w} \in \tilde{S}_n$ in 1-line notation as a permutation of $\mathbb{Z}$.

Generators transpose **infinitely many** pairs of entries:

$$s_i : (\mathbf{i}) \leftrightarrow (\mathbf{i+1}) \dots (n+i) \leftrightarrow (n+i+1) \dots (-n+i) \leftrightarrow (-n+i+1) \dots$$

In $\tilde{S}_4$,	$\dots w(-4)$	$w(-3) \ w(-2) \ w(-1) \ w(0)$	$w(1) \ w(2) \ w(3) \ w(4)$	$w(5) \ w(6) \ w(7) \ w(8)$	$w(9) \dots$
s_1	$\dots -4$	$-2 \ -3 \ -1 \ 0$	$2 \ 1 \ 3 \ 4$	$6 \ 5 \ 7 \ 8$	$10 \dots$
s_0	$\dots -3$	$-4 \ -2 \ -1 \ 1$	$0 \ 2 \ 3 \ 5$	$4 \ 6 \ 7 \ 9$	$8 \dots$

Window notation

Affine n -Permutations $\tilde{S}_n$ (G. Lusztig 1983, H. Eriksson, 1994)

Write an element $\tilde{w} \in \tilde{S}_n$ in 1-line notation as a permutation of $\mathbb{Z}$.

Generators transpose **infinitely many** pairs of entries:

$$s_i : (\mathbf{i}) \leftrightarrow (\mathbf{i+1}) \dots (n+i) \leftrightarrow (n+i+1) \dots (-n+i) \leftrightarrow (-n+i+1) \dots$$

In $\tilde{S}_4$,	$\dots w(-4)$	$w(-3) \ w(-2) \ w(-1) \ w(0)$	$w(1) \ w(2) \ w(3) \ w(4)$	$w(5) \ w(6) \ w(7) \ w(8)$	$w(9) \dots$
s_1	$\dots -4$	$-2 \ -3 \ -1 \ 0$	$2 \ 1 \ 3 \ 4$	$6 \ 5 \ 7 \ 8$	$10 \dots$
s_0	$\dots -3$	$-4 \ -2 \ -1 \ 1$	$0 \ 2 \ 3 \ 5$	$4 \ 6 \ 7 \ 9$	$8 \dots$
$s_1 s_0$	$\dots -2$	$-4 \ -3 \ -1 \ 2$	$0 \ 1 \ 3 \ 6$	$4 \ 5 \ 7 \ 10$	$8 \dots$

Window notation

Affine n -Permutations $\tilde{S}_n$ (G. Lusztig 1983, H. Eriksson, 1994)

Write an element $\tilde{w} \in \tilde{S}_n$ in 1-line notation as a permutation of $\mathbb{Z}$.

Generators transpose **infinitely many** pairs of entries:

$$s_i : (\mathbf{i}) \leftrightarrow (\mathbf{i+1}) \dots (n+i) \leftrightarrow (n+i+1) \dots (-n+i) \leftrightarrow (-n+i+1) \dots$$

In $\tilde{S}_4$,	$\dots w(-4)$	$w(-3) w(-2) w(-1) w(0)$	$w(1) w(2) w(3) w(4)$	$w(5) w(6) w(7) w(8)$	$w(9) \dots$
s_1	$\dots -4$	$-2 -3 -1 0$	$2 1 3 4$	$6 5 7 8$	$10 \dots$
s_0	$\dots -3$	$-4 -2 -1 1$	$0 2 3 5$	$4 6 7 9$	$8 \dots$
$s_1 s_0$	$\dots -2$	$-4 -3 -1 2$	$0 1 3 6$	$4 5 7 10$	$8 \dots$

Symmetry: Can think of as integers wrapped around a cylinder.

Window notation

Affine n -Permutations $\tilde{S}_n$ (G. Lusztig 1983, H. Eriksson, 1994)

Write an element $\tilde{w} \in \tilde{S}_n$ in 1-line notation as a permutation of $\mathbb{Z}$.

Generators transpose **infinitely many** pairs of entries:

$$s_i : (\mathbf{i}) \leftrightarrow (\mathbf{i+1}) \dots (n+i) \leftrightarrow (n+i+1) \dots (-n+i) \leftrightarrow (-n+i+1) \dots$$

In $\tilde{S}_4$,	$\dots w(-4)$	$w(-3) \ w(-2) \ w(-1) \ w(0)$	$w(1) \ w(2) \ w(3) \ w(4)$	$w(5) \ w(6) \ w(7) \ w(8)$	$w(9) \dots$
s_1	$\dots -4$	$-2 \ -3 \ -1 \ 0$	$2 \ 1 \ 3 \ 4$	$6 \ 5 \ 7 \ 8$	$10 \dots$
s_0	$\dots -3$	$-4 \ -2 \ -1 \ 1$	$0 \ 2 \ 3 \ 5$	$4 \ 6 \ 7 \ 9$	$8 \dots$
$s_1 s_0$	$\dots -2$	$-4 \ -3 \ -1 \ 2$	$0 \ 1 \ 3 \ 6$	$4 \ 5 \ 7 \ 10$	$8 \dots$

Symmetry: Can think of as integers wrapped around a cylinder.

$\tilde{w}$ is defined by the window $[\tilde{w}(1), \tilde{w}(2), \dots, \tilde{w}(n)]$. $s_1 s_0 = [0, 1, 3, 6]$

An abacus model for affine permutations

(James and Kerber, 1981) Given an affine permutation $[w_1, \dots, w_n]$,

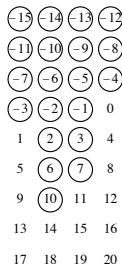
- Place integers in n *runners*.

-15	-14	-13	-12
-11	-10	-9	-8
-7	-6	-5	-4
-3	-2	-1	0
1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16
17	18	19	20

An abacus model for affine permutations

(James and Kerber, 1981) Given an affine permutation $[w_1, \dots, w_n]$,

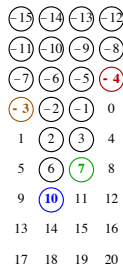
- Place integers in n *runners*.
- Circled: *beads*. Empty: *gaps*



An abacus model for affine permutations

(James and Kerber, 1981) Given an affine permutation $[w_1, \dots, w_n]$,

- Place integers in n *runners*.
- Circled: *beads*. Empty: *gaps*
- Create an abacus where each runner has a lowest bead at w_i .

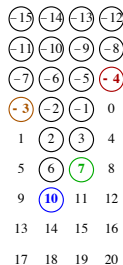


Example: $[-4, -3, 7, 10]$

An abacus model for affine permutations

(James and Kerber, 1981) Given an affine permutation $[w_1, \dots, w_n]$,

- ▶ Place integers in n *runners*.
- ▶ Circled: *beads*. Empty: *gaps*
- ▶ Create an abacus where each runner has a lowest bead at w_i .



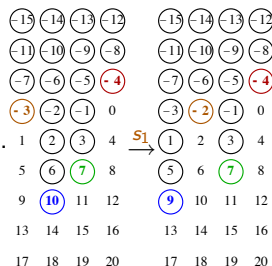
Example: $[-4, -3, 7, 10]$

- ▶ Generators act nicely.
- ▶ s_i interchanges runners $i \leftrightarrow i + 1$.
- ▶ s_0 interchanges runners 1 and n (with shifts)

An abacus model for affine permutations

(James and Kerber, 1981) Given an affine permutation $[w_1, \dots, w_n]$,

- ▶ Place integers in n *runners*.
- ▶ Circled: *beads*. Empty: *gaps*
- ▶ Create an abacus where each runner has a lowest bead at w_i .



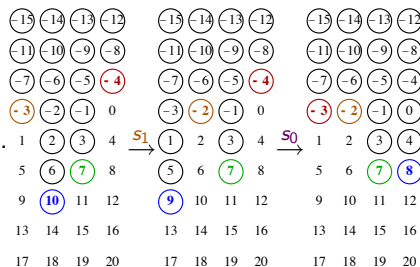
Example: $[-4, -3, 7, 10]$

- ▶ Generators act nicely.
- ▶ s_i interchanges runners $i \leftrightarrow i + 1$. ($s_1 : 1 \leftrightarrow 2$)
- ▶ s_0 interchanges runners 1 and n (with shifts)

An abacus model for affine permutations

(James and Kerber, 1981) Given an affine permutation $[w_1, \dots, w_n]$,

- Place integers in n *runners*.
- Circled: *beads*. Empty: *gaps*
- Create an abacus where each runner has a lowest bead at w_i .

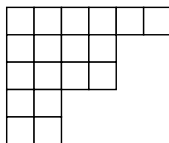


Example: $[-4, -3, 7, 10]$

- Generators act nicely.
- s_i interchanges runners $i \leftrightarrow i + 1$. ($s_1 : 1 \leftrightarrow 2$)
- s_0 interchanges runners 1 and n (with shifts) ($s_0 : 1 \overset{\text{shift}}{\leftrightarrow} 4$)

Core partitions

For an integer partition $\lambda = (\lambda_1, \dots, \lambda_k)$ drawn as a Young diagram,

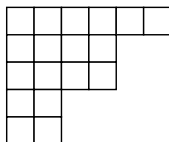


The **hook length** of a box is # boxes below and to the right.

10	9	6	5	2	1
7	6	3	2		
6	5	2	1		
3	2				
2	1				

Core partitions

For an integer partition $\lambda = (\lambda_1, \dots, \lambda_k)$ drawn as a Young diagram,



The **hook length** of a box is # boxes below and to the right.

10	9	6	5	2	1
7	6	3	2		
6	5	2	1		
3	2				
2	1				

An **n -core** is a partition with no boxes of hook length dividing n .

Example. λ is a 4-core, 8-core, 11-core, 12-core, etc.

λ is NOT a 1-, 2-, 3-, 5-, 6-, 7-, 9-, or 10-core.

Core partition interpretation for affine permutations

Bijection: $\{\text{abaci}\} \longleftrightarrow \{n\text{-cores}\}$

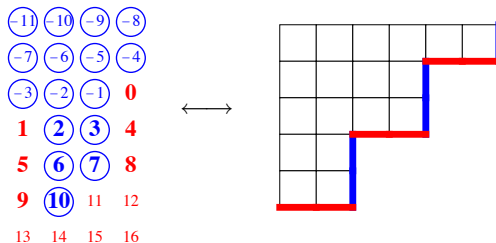
Core partition interpretation for affine permutations

Bijection: $\{\text{abaci}\} \longleftrightarrow \{n\text{-cores}\}$

Rule: Read the boundary steps of λ from the abacus:

► A bead $\leftrightarrow$ vertical step

► A gap $\leftrightarrow$ horizontal step



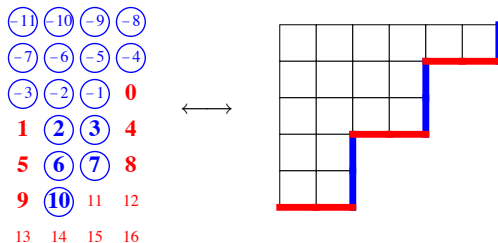
Core partition interpretation for affine permutations

Bijection: $\{\text{abaci}\} \longleftrightarrow \{n\text{-cores}\}$

Rule: Read the boundary steps of λ from the abacus:

► A bead $\leftrightarrow$ vertical step

► A gap $\leftrightarrow$ horizontal step



Fact: This is a bijection!

Action of generators on the core partition

- ▶ Label the boxes of λ with residues.
- ▶ s_i acts by adding or removing boxes with residue i .

0	1	2	3	0	1
3	0	1	2	3	0
2	3	0	1	2	3
1	2	3	0	1	2
0	1	2	3	0	1
3	0	1	2	3	0

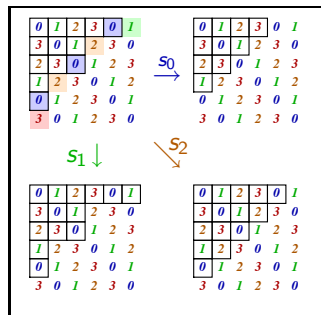
Action of generators on the core partition

- ▶ Label the boxes of λ with residues.
- ▶ s_i acts by adding or removing boxes with residue i .

0	1	2	3	0	1
3	0	1	2	3	0
2	3	0	1	2	3
1	2	3	0	1	2
0	1	2	3	0	1
3	0	1	2	3	0

Example. $\lambda = (5, 3, 3, 1, 1)$

- ▶ has removable 0 boxes
- ▶ has addable 1, 2, 3 boxes.



Action of generators on the core partition

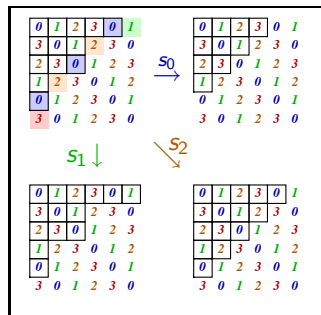
- ▶ Label the boxes of λ with residues.
- ▶ s_i acts by adding or removing boxes with residue i .

0	1	2	3	0	1
3	0	1	2	3	0
2	3	0	1	2	3
1	2	3	0	1	2
0	1	2	3	0	1
3	0	1	2	3	0

Example. $\lambda = (5, 3, 3, 1, 1)$

- ▶ has removable 0 boxes
- ▶ has addable 1, 2, 3 boxes.

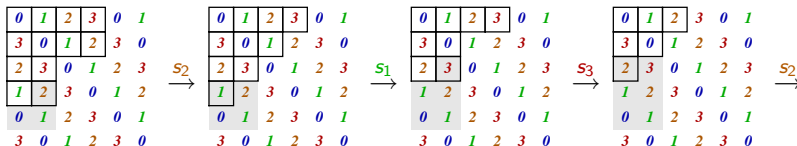
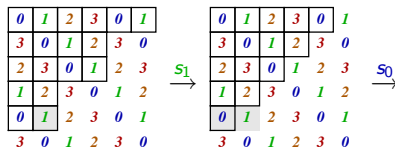
Idea: We can use this to figure out a *word* for w .



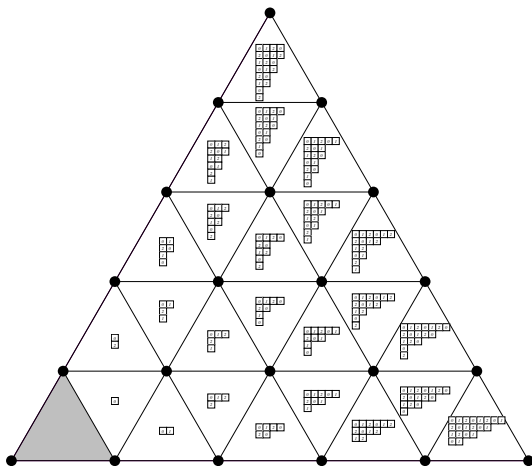
Finding a word for an affine permutation.

Example: The word in S_4 corresponding to $\lambda = (6, 4, 4, 2, 2)$:

$s_1 s_0 s_2 s_1 s_3 s_2 s_0 s_3 s_1 s_0$

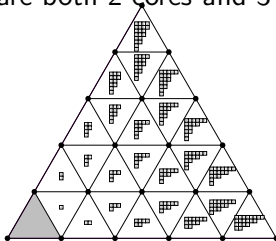


The bijection between cores and alcoves



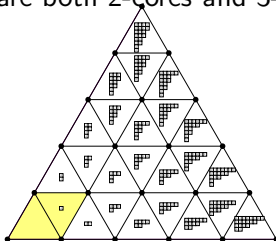
Simultaneous core partitions

How many partitions are both 2-cores and 3-cores?



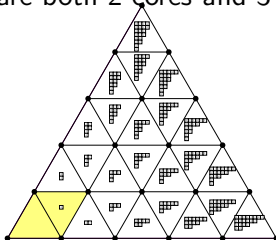
Simultaneous core partitions

How many partitions are both 2-cores and 3-cores? **2.**



Simultaneous core partitions

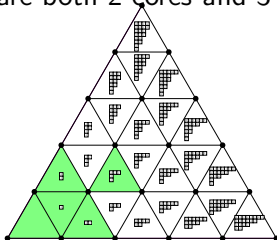
How many partitions are both 2-cores and 3-cores? **2.**



How many partitions are both 3-cores and 4-cores?

Simultaneous core partitions

How many partitions are both 2-cores and 3-cores? **2.**

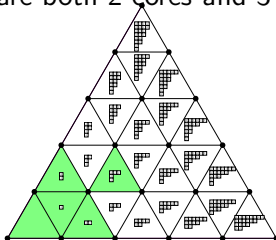


How many partitions are both 3-cores and 4-cores? **5.**

How many simultaneous 4/5-cores?

Simultaneous core partitions

How many partitions are both 2-cores and 3-cores? **2.**

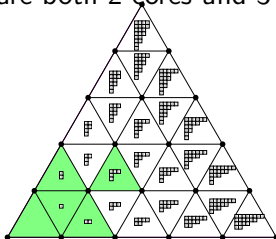


How many partitions are both 3-cores and 4-cores? **5.**

How many simultaneous 4/5-cores? **14.**

Simultaneous core partitions

How many partitions are both 2-cores and 3-cores? **2.**



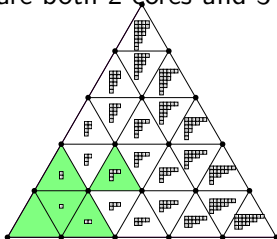
How many partitions are both 3-cores and 4-cores? **5.**

How many simultaneous 4/5-cores? **14.**

How many simultaneous 5/6-cores? **42.**

Simultaneous core partitions

How many partitions are both 2-cores and 3-cores? **2.**



How many partitions are both 3-cores and 4-cores? **5.**

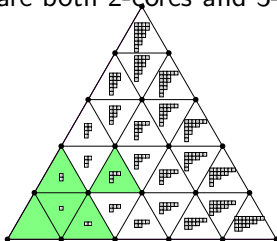
How many simultaneous 4/5-cores? **14.**

How many simultaneous 5/6-cores? **42.**

How many simultaneous $n/(n+1)$ -cores?

Simultaneous core partitions

How many partitions are both 2-cores and 3-cores? **2.**



How many partitions are both 3-cores and 4-cores? **5.**

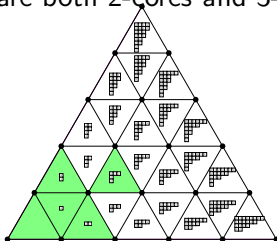
How many simultaneous 4/5-cores? **14.**

How many simultaneous 5/6-cores? **42.**

How many simultaneous $n/(n+1)$ -cores? **$C_n!$**

Simultaneous core partitions

How many partitions are both 2-cores and 3-cores? **2.**



How many partitions are both 3-cores and 4-cores? **5.**

How many simultaneous 4/5-cores? **14.**

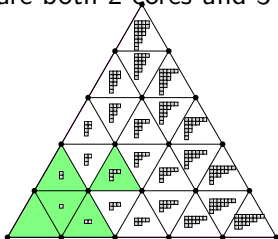
How many simultaneous 5/6-cores? **42.**

How many simultaneous $n/(n+1)$ -cores? **$C_n!$**

Jaclyn Anderson proved that the number of s/t -cores is $\frac{1}{s+t} \binom{s+t}{s}$.

Simultaneous core partitions

How many partitions are both 2-cores and 3-cores? **2.**



How many partitions are both 3-cores and 4-cores? **5.**

How many simultaneous 4/5-cores? **14.**

How many simultaneous 5/6-cores? **42.**

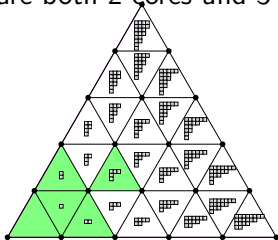
How many simultaneous $n/(n+1)$ -cores? **$C_n!$**

Jaclyn Anderson proved that the number of s/t -cores is $\frac{1}{s+t} \binom{s+t}{s}$.

The number of 3/7-cores is $\frac{1}{10} \binom{10}{3} = \frac{1}{10} \frac{10 \cdot 9 \cdot 8}{3 \cdot 2 \cdot 1} = 12$.

Simultaneous core partitions

How many partitions are both 2-cores and 3-cores? **2.**



How many partitions are both 3-cores and 4-cores? **5.**

How many simultaneous 4/5-cores? **14.**

How many simultaneous 5/6-cores? **42.**

How many simultaneous $n/(n+1)$ -cores? **$C_n!$**

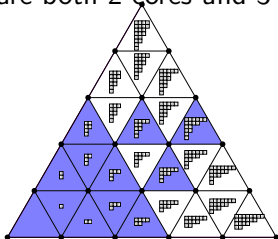
Jaclyn Anderson proved that the number of s/t -cores is $\frac{1}{s+t} \binom{s+t}{s}$.

The number of $3/7$ -cores is $\frac{1}{10} \binom{10}{3} = \frac{1}{10} \frac{10 \cdot 9 \cdot 8}{3 \cdot 2 \cdot 1} = 12$.

Fishel–Vazirani proved an alcove interpretation of $n/(mn+1)$ -cores. ↻ 🔍

Simultaneous core partitions

How many partitions are both 2-cores and 3-cores? **2.**



How many partitions are both 3-cores and 4-cores? **5.**

How many simultaneous 4/5-cores? **14.**

How many simultaneous 5/6-cores? **42.**

How many simultaneous $n/(n+1)$ -cores? $C_n!$

Jaclyn Anderson proved that the number of s/t -cores is $\frac{1}{s+t} \binom{s+t}{s}$.

The number of $3/7$ -cores is $\frac{1}{10} \binom{10}{3} = \frac{1}{10} \frac{10 \cdot 9 \cdot 8}{3 \cdot 2 \cdot 1} = 12$.

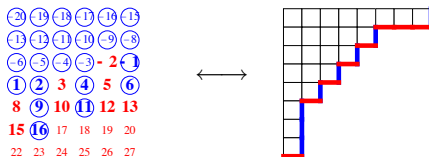
Fishel–Vazirani proved an alcove interpretation of $n/(mn+1)$ -cores. ↻ 🔍

Research Questions

★ Can we extend combinatorial interps to other reflection groups?

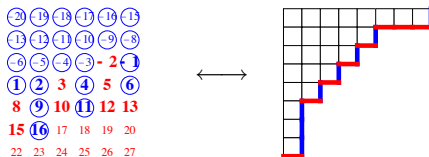
Research Questions

- ★ Can we extend combinatorial interps to other reflection groups?
 - Yes! Involves self-conjugate partitions.



Research Questions

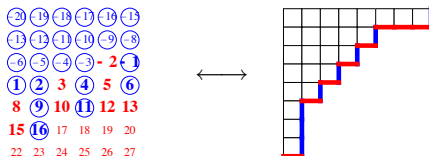
- ★ Can we extend combinatorial interps to other reflection groups?
 - ▶ Yes! Involves self-conjugate partitions. [arXiv:1105.5333](#)
 - ▶ Joint with Brant Jones, James Madison University.



- ★ What numerical properties do self-conjugate core partitions have?
 - ▶ Joint with Rishi Nath, York College. [arXiv:1201.6629](#)

Research Questions

- ★ Can we extend combinatorial interps to other reflection groups?
 - ▶ Yes! Involves self-conjugate partitions. [arXiv:1105.5333](#)
 - ▶ Joint with Brant Jones, James Madison University.

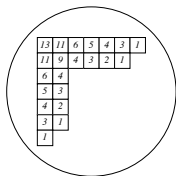


- ★ What numerical properties do self-conjugate core partitions have?
 - ▶ Joint with Rishi Nath, York College. [arXiv:1201.6629](#)
 - ▶ We found & proved some impressive numerical conjectures.

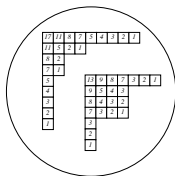
Research Questions

- ★ Can we extend combinatorial interps to other reflection groups?
 - ▶ Yes! Involves self-conjugate partitions. [arXiv:1105.5333](#)
 - ▶ Joint with Brant Jones, James Madison University.
- ★ What numerical properties do self-conjugate core partitions have?
 - ▶ Joint with Rishi Nath, York College. [arXiv:1201.6629](#)
 - ▶ We found & proved some impressive numerical conjectures.
 - ▶ There are more (s.c. $t+2$ -cores of n) than (s.c. t -cores of n).

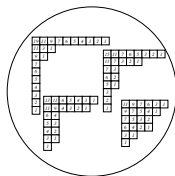
4-cores of 22



6-cores of 22



8-cores of 22



Research Questions

- ★ Can we extend combinatorial interps to other reflection groups?
 - ▶ Yes! Involves self-conjugate partitions. [arXiv:1105.5333](#)
 - ▶ Joint with Brant Jones, James Madison University.
- ★ What numerical properties do self-conjugate core partitions have?
 - ▶ Joint with Rishi Nath, York College. [arXiv:1201.6629](#)
 - ▶ We found & proved some impressive numerical conjectures.
 - ▶ There are more (s.c. $t+2$ -cores of n) than (s.c. t -cores of n).
- ★ What is the average size of an s/t -core partition?
 - ▶ In progress. We “know” the answer, but we have to prove it!

Research Questions

- ★ Can we extend combinatorial interps to other reflection groups?
 - ▶ Yes! Involves self-conjugate partitions. [arXiv:1105.5333](#)
 - ▶ Joint with Brant Jones, James Madison University.
- ★ What numerical properties do self-conjugate core partitions have?
 - ▶ Joint with Rishi Nath, York College. [arXiv:1201.6629](#)
 - ▶ We found & proved some impressive numerical conjectures.
 - ▶ There are more (s.c. $t+2$ -cores of n) than (s.c. t -cores of n).
- ★ What is the average size of an s/t -core partition?
 - ▶ In progress. We “know” the answer, but we have to prove it!
 - ▶ Working with Drew Armstrong, University of Miami.

Thank you!

Slides available: people.qc.cuny.edu/chanusa > Talks

Interact: people.qc.cuny.edu/chanusa > Animations



M. A. Armstrong.

Groups and symmetry. Springer, 1988.

Easy-to-read introduction to groups, (esp. reflection)



James E. Humphreys

Reflection groups and Coxeter groups. Cambridge, 1990.

More advanced and the reference for reflection groups.



<http://www.mcescher.com/>



<http://www.math.ubc.ca/~cass/coxeter/crm1.html>



<http://sourceforge.net/apps/trac/groupexplorer/wiki/>

The First Five Symmetric Groups/

